

Session 1. Predict

Workshop Exercises

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SESSION 1

Running jobs on the AlphaFold Web Server and interpreting confidence metrics

Rice immune receptor Pikp-1 causes resistance to *Pyricularia oryzae*

Rice blast disease caused by *Pyricularia oryzae*



Pyricularia oryzae
spores



- *Pikp-1*



+ *Pikp-1*

Rice cultivars



Protein sequence of Pikip-1

MEAAAMAVTAATGALAPVLVKLAALLDDGECNLLEGSRSDAEFIRSELEAVHSLLTPNILGRMGD
DDAACKDGLIAEVRELSYDLDDAVDDFLELNFEEQRRSASPFGELEKARVEEHVSNRFSWKLPAAS
LPPSSVHRRAGLPPPDALVGMCKRMEELTKLLEQGSNDASRWRKRKPHFPLRKTGLKQKIVIKV
AMEGNNCRSKAMALVASTGGVDSVALVGDLRDKIEVVGYGIDPIKLISALRKKVGDAELLQVSQA
NKDVKETTPMLAPVKSICEFHVKVTVCILGLPGGGKTTVARELYDALGTHFPCRVFVSVSPSSSP
SPNLTKTLADIFAQAQLGVTDLTSTSYGGSGTGRALQQHLIDNISAFLLNKKYLIVIDDIWHWEE
WEVIRKSIPKNDLGGRIIMTTRLNSIAEKCHTDDNDVFVYEVGDLNDNDALSLSWGIATKSGAGN
RIGTGEDNPCYDIVNMCYGMPLALIWLSSALVGEIEELGGAEVKKCRDLRHIEDGILDIPSLQPL
AESLCLGYNHLPLYLRLLLYCSAYHWSNRIERGRLVRRWIAEGFVSEEKEAEGYFGELINRGWI
TQHGDNNSYNYYEIHVPMLAFLRCKSKEYNFLTCLGLGSDTSTSASSPRLIRRLSLQGGYPVDCL
SSMSMDVSHTCSLVVLGDVARPKGIPFYMFKRLRVLDLEDNKDIQDSHLQGICEQLSLRVRYLGL
KGTRIRKLPQEMRKLKHLEILYVGSTRISELPQEIGELKHLRILDVRNTDITELPLQIRELQHLH
TLDVRNTPISELPPQVGKLQNLKIMCVRSTGVRELPKEIGELNHLQTL DVRNTRVRELPWQAGQI
SQSLRVLAGDSGDGVRLPEGVCEALINGIPGATRAKCREVLSIAIIDRFGPPLVGIFKVPGSHMR
IPKMIKDHFRVLSCLDIRLCHKLEDDDDQKFLAEMPNLQTLVLRFEALPRQPITINGTGFQMLESF
RVDSRVPRIAFHEDAMPNLKLLLEFKFYAGPASNDAIGITNLKSLQKVVFRCSPWYKSDAPGISAT
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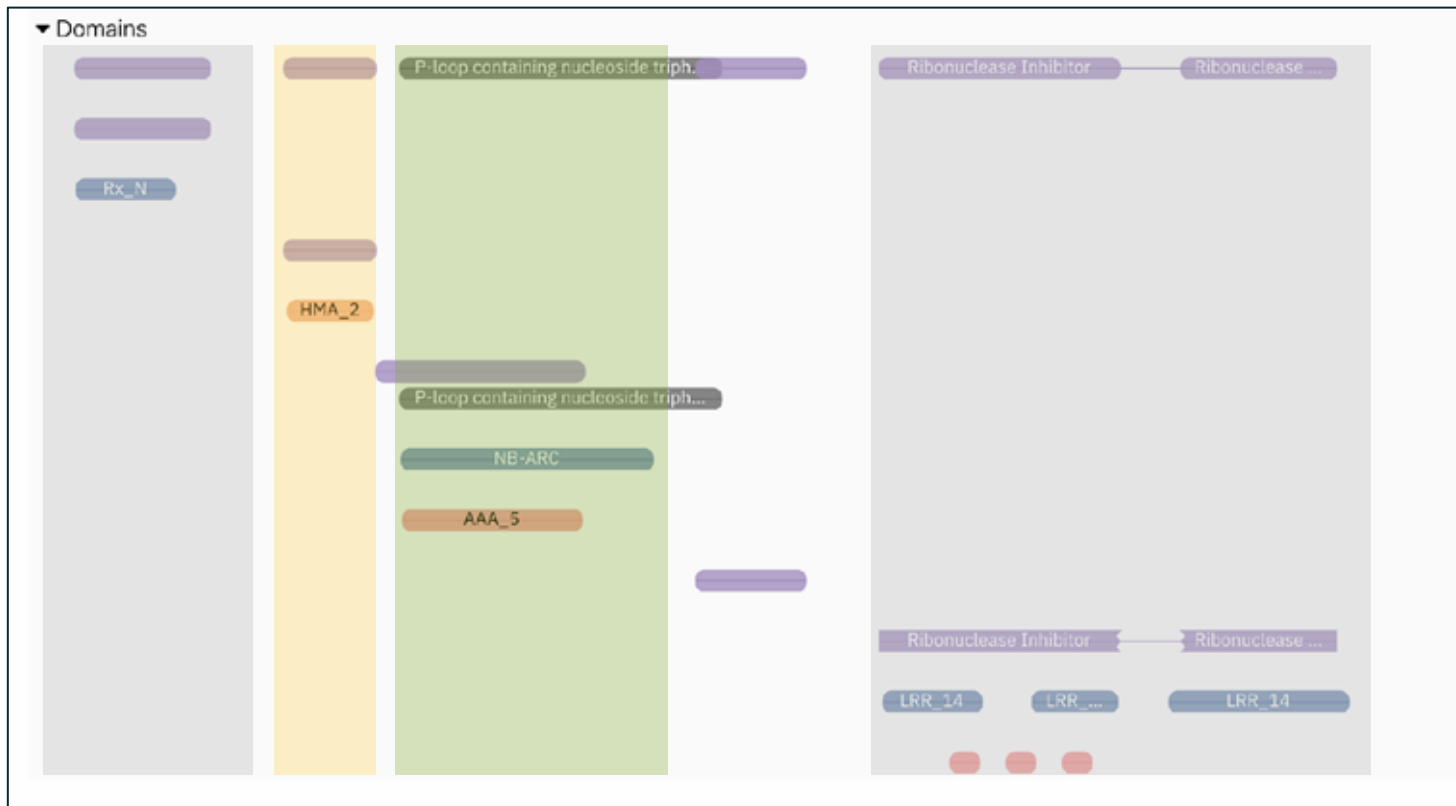
Protein sequence of Pikip-1

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SPNLTKTLADIFAQAQLGVTDTLSTSYGGSGTGRALQQHLIDNISAFLLNKKYLIVIDDIWHWEE
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SQSLRVLAGDSGDGVRLPEGVCEALINGIPGATRAKCREVLSIAIIDRFGPPLVGIFKVP GSHMR
IPKMIKDHFRVLSCLDIRLCHKLEDDDDQKFLAEMP NLQTLVLRFEALPRQPITINGTGFQMLESF
RVDSRVPRIAFHEDAMPNLKLEFKFYAGPASNDAIGITNLKSLQKVVFRCSPWYKSDAPGISAT
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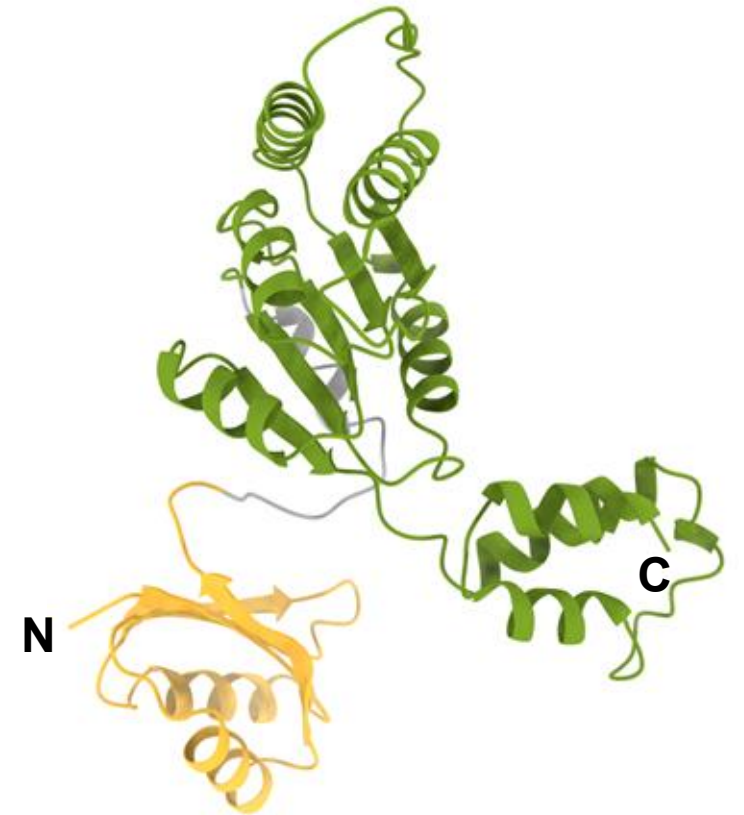
Residues 186–486

Domain architecture of Pikp-1

Residues 186–486



Domain annotation from InterProScan



EXERCISE 1

Modelling individual structures of proteins

In this exercise, you will model the structure of the partial sequence of the rice immune receptor protein, **Pikp-1**.

To begin, navigate to <https://alphafoldserver.com/>, create an account, and submit your first job:

>Pikp-1_186-486

```
GLKQKIVIKVAMEGNNCRSKAMALVASTGGVDSVALVGDLRDKIEVVGYGIDPIKLISALRKKVGDAELLQVSQANKDVKETTPMLAPVKSICEFHKVKTVCILGLPGGGKTTVARELYDAL  
GTHFPCR VFVSVPSSSPSPNLTKTLADIFAQAQLGVTDTLSTSYGGSGTGRALQQHLIDNISAFLLNKKYLIVIDDIWHWEWEVIRKSI PKNDLGGRI IMTTRLNSIAEKCHTDDNDVFV  
YEVGDLNNDALSLSWGIATKSGAGNRIGTGEDNPCYDIVNMCYGMPLALIWLSSAL
```

Modelling individual structures of proteins

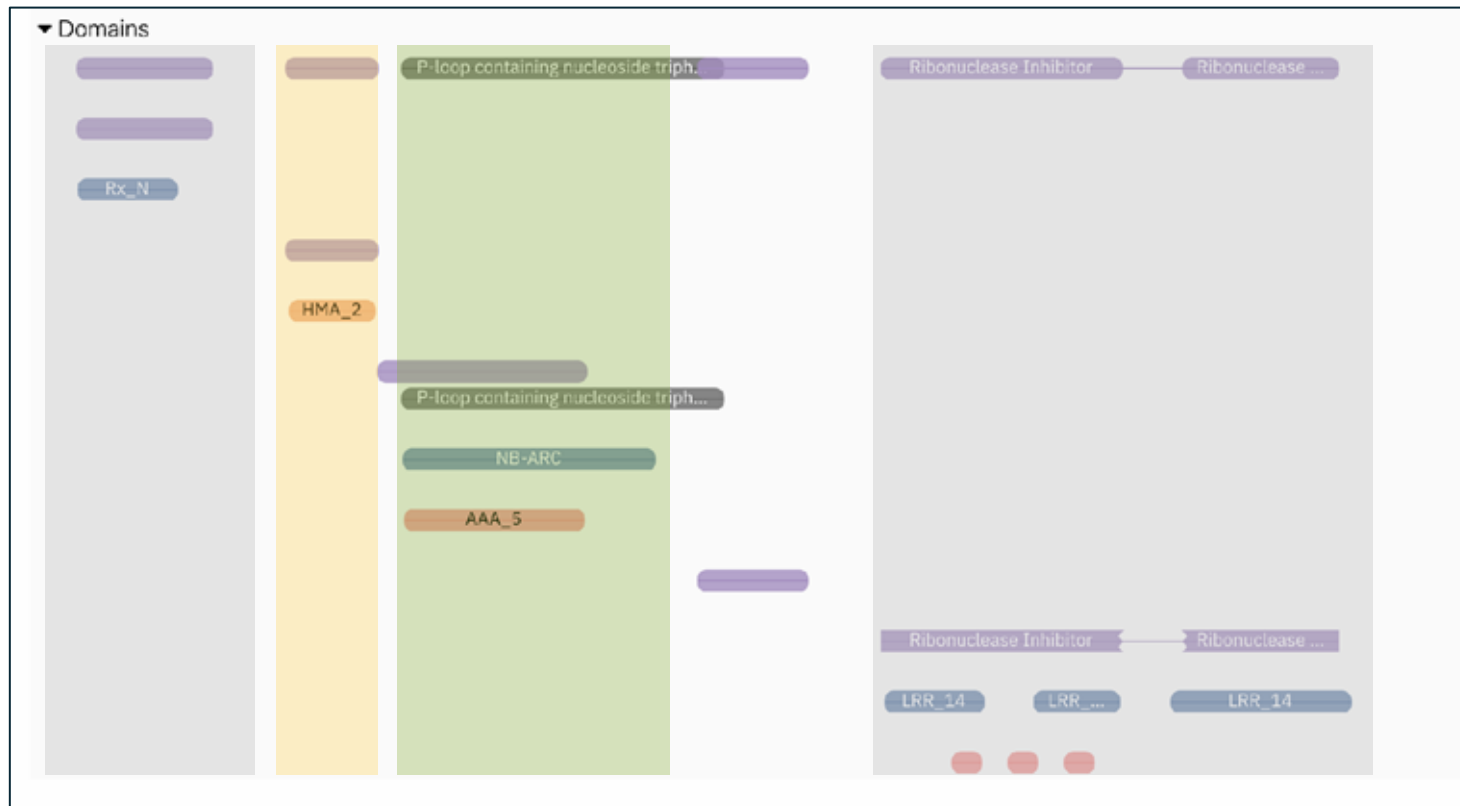


DISCUSSION

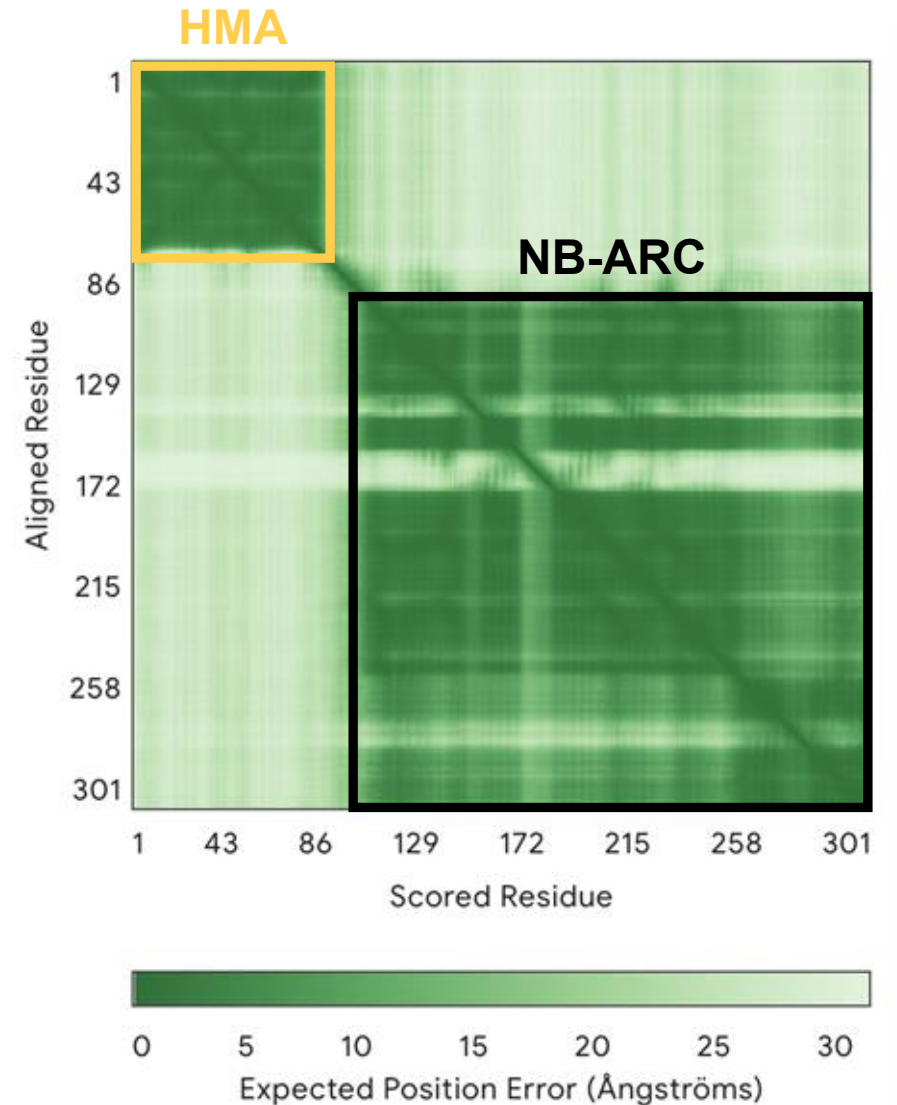
- a) What does pTM stand for? What is the pTM score for the prediction?
- b) What can you say about the local measure of confidence (pLDDT)? Is it high all over the model? Are there any amino acids predicted with higher confidence than others?
- c) What can you say about the predicted aligned error (PAE) plot? Are there any regions where the relative positions are unreliable?

Domain architecture and PAE plot of Pikip-1

Residues 186–486



Domain annotation from InterProScan



EXERCISE 2

Modelling a multimeric complex

In this exercise, you will explore the oligomeric state predictions for a given protein. You will model the HMA domain of the Pkp-1 protein (from Exercise 1.1) in **dimeric (n=2)**, **trimeric (n=3)**, and **tetrameric (n=4)** states.

Your task is to evaluate the confidence in the relative orientation of the modeled polypeptide chains by interpreting the PAE plots. You will complete the table with predicted pTM and ipTM scores and, based on this, determine the likely oligomeric state (monomer, dimer, trimer, or tetramer).

>Pkp-HMA

GLKQKIVIKVAMEGNNCRSKAMALVASTGGVDSVALVGDLRDKIEVVGYGIDPIKLI SALRKKVGD AELLQVSQANKD

EXERCISE 2

Complete the results table

a) What does ipTM stand for?

b) Complete the table below with your modelling results:

Oligomeric state	ipTM	pTM
Monomer (n = 1)	—	
Dimer (n = 2)		
Trimer (n = 3)		
Tetramer (n = 4)		

Fill in during the session as you run each stoichiometry (n = 1 to 4).

Interpreting the PAE across oligomeric states

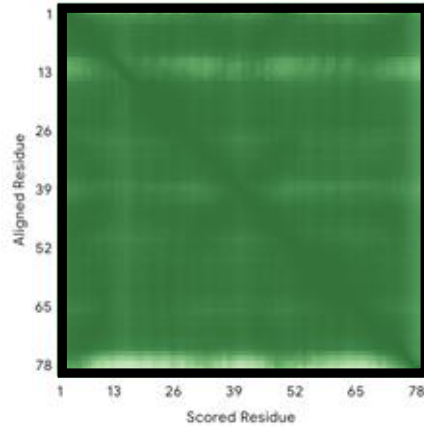


DISCUSSION

- c) What does the predicted alignment error (PAE) look like for the different oligomeric states? Is there high confidence in the relative positioning of the protomers, across all cases tested?

Pikp-HMA shows the highest ipTM score as a homodimer

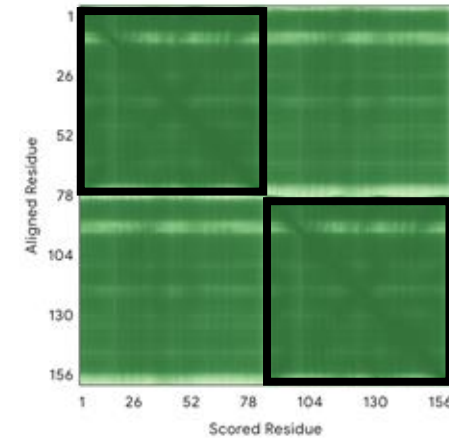
Monomer (n=1)



pTM: 0.81

ipTM: N/A

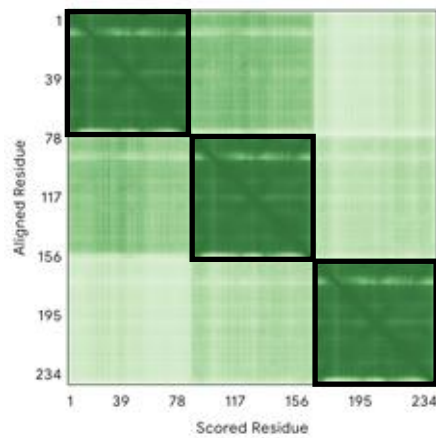
Dimer (n=2)



pTM: 0.86

ipTM: 0.83

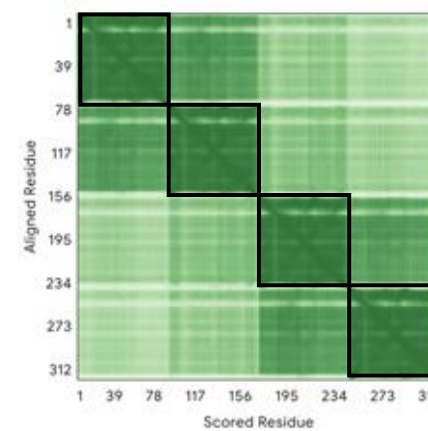
Trimer (n=3)



pTM: 0.49

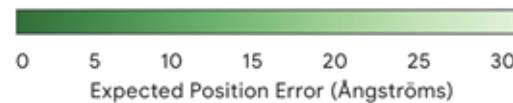
ipTM: 0.3

Tetramer (n=4)

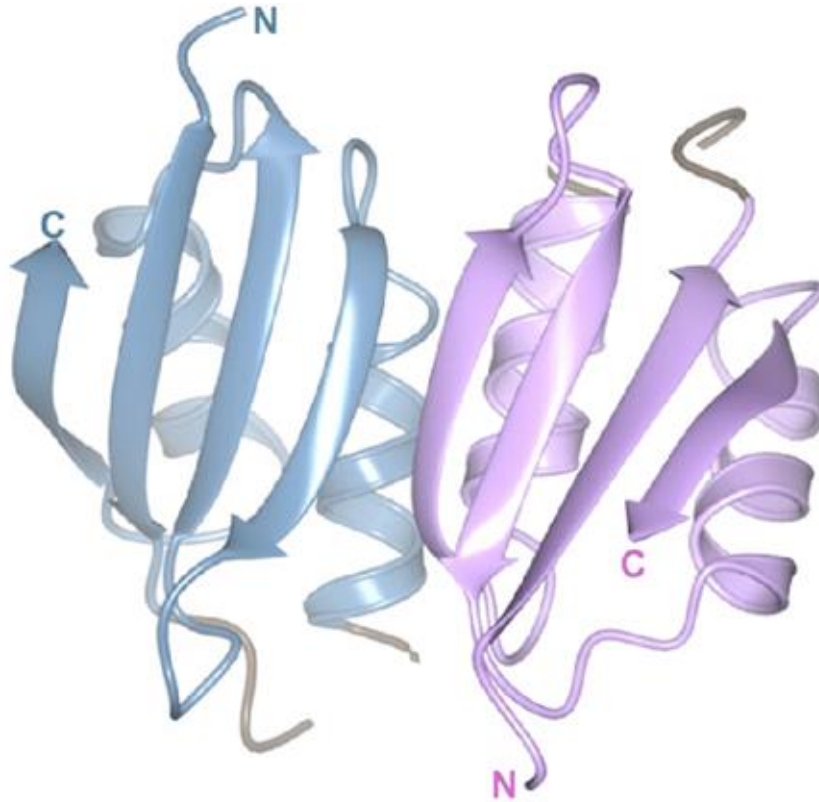


pTM: 0.59

ipTM: 0.49



Crystal structure of Pkcp-HMA has been solved as a homo-dimer



PDB: 5a6p

(Maqbool et al., 2015, PMID: 26304198)

EXERCISE 3

Protein–protein interaction prediction

Next, model three different heteromeric complexes between pathogen proteins and the rice Pikp-1 domains. Your goal is to analyze the AF3 models and make a hypothesis about **which effector may interact with the plant host protein**, based on the modelling confidence metrics alone.

>Pikp-1_186-486

```
GLKQKIVIKVAMEGNNCRSKAMALVASTGGVDSVALVGDLRDKIEVVGYGIDPIKLISALRKKVGDAELLQVSQANKDVKETTPMLAPVKSICEFHKVKTVCILGLPGGGKTTVARELYDAL  
GTHFPCR VFVSVPSSSPNLTKTLADIFAQAQLGVTDTLSTSYGGSGTGRALQQHLIDNISAFLLNKKYLIVIDDIWHWEWEVIRKSI PKNDLGGRI IMTTRLNSIAEKCHTDDNDVFV  
YEVGDLNNDALSLSWG IATKSGAGNRIGTGEDNPCYDIVNMCYGMPLALIWLSSAL
```

EXERCISE 3

Three pathogen effector candidates

Amino acid sequences of three pathogen proteins secreted by the fungus *Pyricularia oryzae* (causal agent of rice blast disease):

>Pathogen_protein_1 (AVR-PikD)

ETGNKYIEKRAIDLSRERDPNFFDHPIPVPECFWFMFKNNVRQDAGTCYSSWKMDMKVGPNWVHIKSDDNCNLSGDFPPGWIVLGKKRPGF

>Pathogen_protein_2 (AVR-Pib)

ATQVTILKKGERITWVEVPKGESREFNIRGKYFTVSVSDDGTPSISGSKYTVE

>Pathogen_protein_3 (AVR-Pii)

LPTPASLNGNTEVATISDVKLEARSDTTYHKCSKCGYGSDSDAYFNHKCN

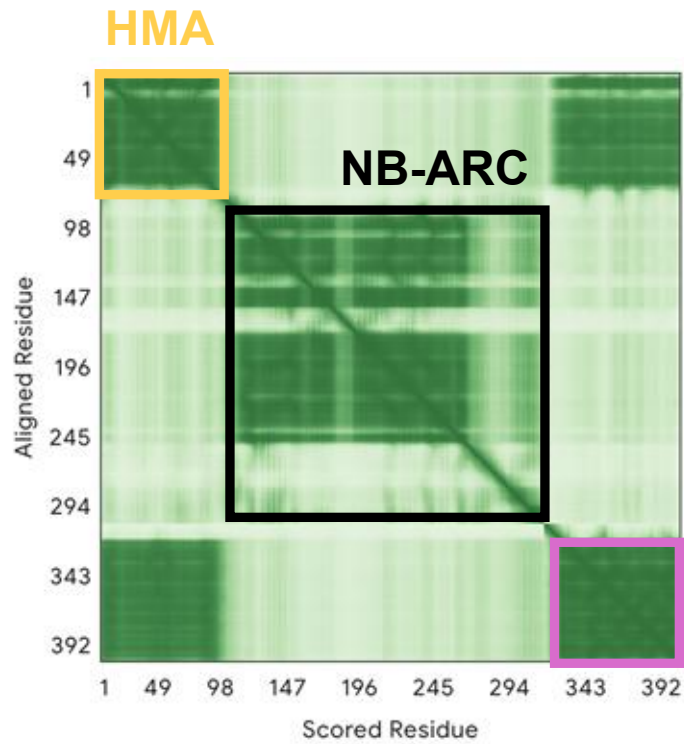
Form and test your hypothesis



DISCUSSION

- a) Based on these modelling results, which pathogen effector(s) is most likely to interact with the rice Pknp-1? What confidence metrics allow you to conclude this?
- b) Based on the modelling results, which Pknp-1 domain contributes to binding to the pathogen protein?
- c) What kind of experiments would you perform to test this hypothesis?

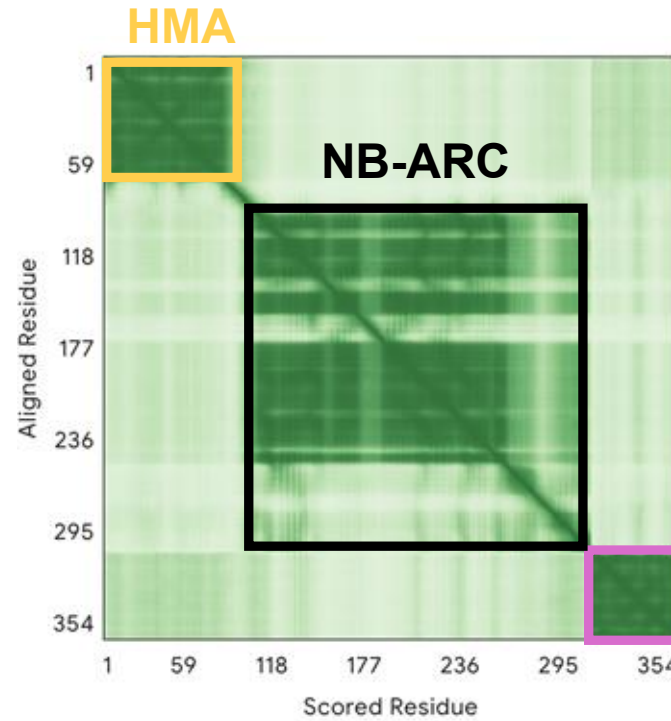
Pathogen_protein_1 shows the highest ipTM score



Pathogen_protein_1
(AVR-PikD)

pTM: 0.45

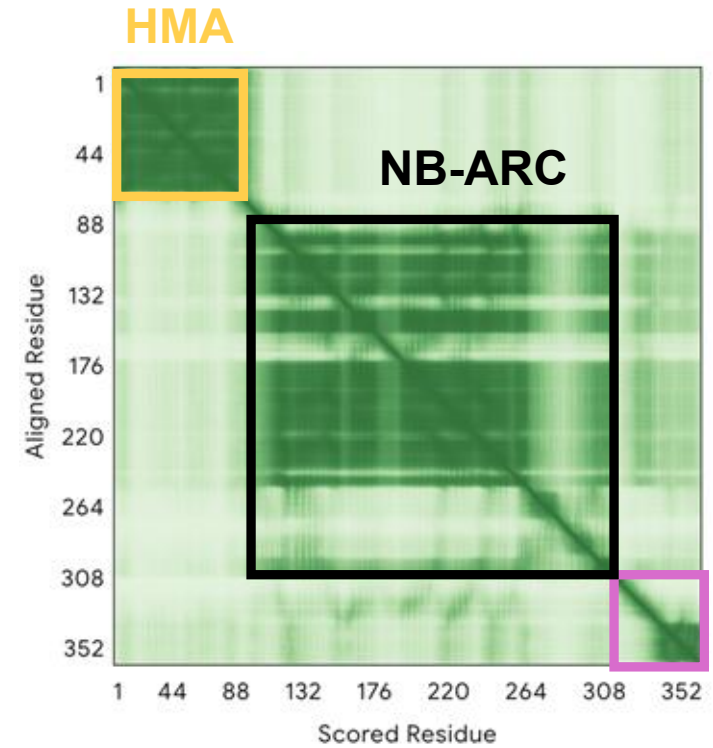
ipTM: 0.83



Pathogen_protein_2
(AVR-Pib)

pTM: 0.47

ipTM: 0.16



Pathogen_protein_3
(AVR-Pii)

pTM: 0.22

ipTM: 0.5